

Sukhrob Ilyosbekov

Machine Learning Engineer · Computer Vision & LLM Systems

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SUMMARY

Machine learning engineer who takes models from research question to production service. Trained in computer vision and multimodal learning, with published work in medical imaging and representation learning, on top of nine years of building software that people depend on. Works across the whole loop: data, training and evaluation, serving, and the monitoring that follows a launch. Applied-math background.

TECHNICAL SKILLS

Deep Learning & CV: PyTorch, TensorFlow, CNNs, Vision Transformers, DINOv3, EfficientNet, YOLO, contrastive learning, GradCAM++, OpenCV

LLM Systems: Claude/GPT/DeepSeek APIs, tool-use agents, MCP, RAG, vision-language models, Hugging Face

Math & Modeling: Probability and statistics, linear algebra, optimization, time-series forecasting, feature engineering, A/B testing, error analysis

Languages: Python, C#, C++, TypeScript, Kotlin

Data & Serving: NumPy, pandas, scikit-learn, FastAPI, PostgreSQL, SQL Server, MongoDB, Redis, OHIF/DICOM

Infrastructure: CUDA, mixed-precision and multi-GPU training, model serving and drift monitoring, Docker, Kubernetes, GitHub Actions, AWS, Azure

Software: ASP.NET Core, SignalR, Blazor, Node.js, Bun, NestJS, React, Next.js, Angular

EDUCATION

Northeastern University, M.S. in Computer Science, GPA 4.0, Boston, MA

Jan 2024 – May 2026

Suffolk University, B.S. in Computer Science, Cum Laude, Boston, MA

Sep 2021 – May 2023

Honors: National Physics Olympiad, 1st Place (Uzbekistan) · Associate Member, Sigma Xi · University Coding Competition Winner, TUIT

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | [EmTech Care Labs](#) | Portland, ME

Jan 2025 – Present

- Built a model that flags Alzheimer's residents at risk of decline from their daily activity and caregiver logs, so staff can intervene before a crisis; 0.81 AUC on held-out data.
- Built an NLP pipeline that reads free-text clinical notes and turns them into structured records of symptoms, medications, and care events, replacing manual chart review; 0.90 F1.
- Put both models into production on de-identified data under HIPAA: nightly retraining, alerts when incoming data drifts, encrypted PHI, and an audit trail for every prediction.
- Led engineering for the care platform itself (TypeScript monorepo on AWS) and took it through an outside HIPAA readiness review with no critical findings.

Machine Learning Engineer | [AllFactors](#) | Santa Clara, CA

Oct 2021 – Dec 2024

- Built property-valuation and market-forecast models trained on 2M listings across 40+ metros, cutting median valuation error from 9% to 5% versus the rules-based system they replaced.
- Built the real-time feature pipeline behind them on SignalR and SQL Server change tracking, so new transactions reach the models within seconds and feed nightly retraining.
- Delivered predictions to users through Blazor WebAssembly dashboards, with automated tests covering the data and prediction flows end to end.

Earlier: Software Engineer, Smart Meal Service, Russia (2020 – 2021), C#/.NET control software for food kiosks and a robotic cashier. Game Developer, Pentallight Technology, Malaysia (2020 – 2021), Unity networking and hand tracking for a VR simulation. Freelance Developer (2019 – 2020), Python NLP and .NET. Game Developer, EC Dev Team, Uzbekistan (2016 – 2019).

SELECTED AI/ML PROJECTS

MorphoCLIP

[GitHub](#) | [Paper](#)

PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1

- Contrastive image-text model matching Cell Painting images of drug- and gene-perturbed cells to plain-language treatment descriptions.
- Frozen DINOv3 and BioClinical ModernBERT backbones with trained projection heads; trains on one consumer GPU.
- Evaluated on CPJUMP1 (51 plates, 3M+ images). First author; led design, training, and evaluation. On arXiv.

Localize, Don't Beautify

<https://arxiv.org/abs/2608.02841>

Image-editing APIs, inpainting, ArcFace, prompt engineering

- Benchmarked prompt steering, masked compositing, and inpainting across six commercial image editors on 196 cosmetic-surgery preview edits.
- Scored identity preservation (ArcFace) and edit localization; masked compositing localized best. Sole author. On arXiv.

MelanomaNet

[GitHub](#) | [Paper](#)

PyTorch, EfficientNet V2, focal loss, GradCAM++, OpenCV, ISIC 2019

- EfficientNet V2 with focal loss over nine ISIC 2019 categories, 25K images; 0.86 weighted F1.
- GradCAM++ attention scored against ABCDE dermatology criteria measured from the lesion mask. Sole author. On arXiv.

Med Image Scanner

<https://github.com/suxrobgm/med-image-scanner>

FastAPI, PyTorch, OpenCV, OHIF Viewer, DICOM, Next.js

- Pulls X-ray, CT, and MRI from hospital imaging systems over DICOM and runs PyTorch detectors: pneumonia on chest X-ray, hemorrhage on head CT.
- Renders predictions as overlays inside the OHIF viewer; on-the-fly de-identification, audit logging, and role-based access.

LogisticsX

<https://logisticsx.app> | [GitHub](#)

Claude API, MCP, .NET 10, EF Core, SignalR, Angular 21, Kotlin Multiplatform, PostgreSQL

- Claude tool-use agent that matches freight loads to trucks, checks federal hours-of-service, and plans multi-stop routes without a dispatcher.
- Full product around it: multi-tenant .NET backend, Angular portals, Kotlin Multiplatform driver app, load-board and telematics integrations.

Bookshelf Scanner

<https://github.com/suxrobgm/bookshelf-scanner>

YOLO, Moondream2 VLM, llama.cpp, FastAPI, Angular

- YOLO segmentation isolates each book spine; Moondream2 reads title and author; Angular UI for corrections and export. Course Outstanding Project Award.

OGStack

<https://ogstack.dev> | [GitHub](#)

Next.js 16, ElysiaJS, Bun, Satori, Cloudflare R2, Stripe

- Multi-model layer (Claude, GPT, DeepSeek) that reads a page's branding and generates its social-share image; 10K+ images in beta, served from a global CDN.